

Question: Explain why inter-VLAN routing is required here, and justify the specific advantages of a Router-on-a-Stick configuration over adding a second physical router interface.

Inter-VLAN Routing

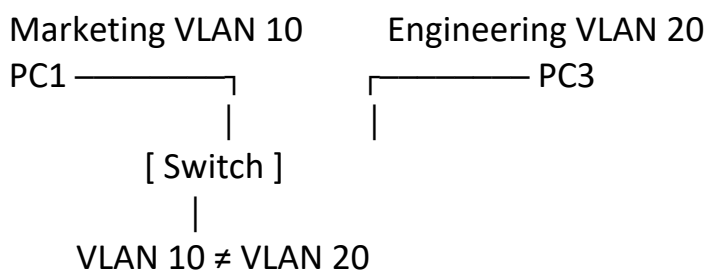
In this scenario, the startup has two VLANs:

- **Marketing VLAN** → e.g. VLAN 10
- **Engineering VLAN** → e.g. VLAN 20
- Both VLANs are connected to the **same switch**
- There is only **one router**

Why is Inter-VLAN Routing required?

By default, **different VLANs are separate Layer-2 broadcast domains.**

For example:



A normal Layer-2 switch can forward frames **within the same VLAN**, but it cannot route traffic from VLAN 10 to VLAN 20.

So if:

PC1 (VLAN 10)
IP: 192.168.10.10
↓
PC3 (VLAN 20)
IP: 192.168.20.10

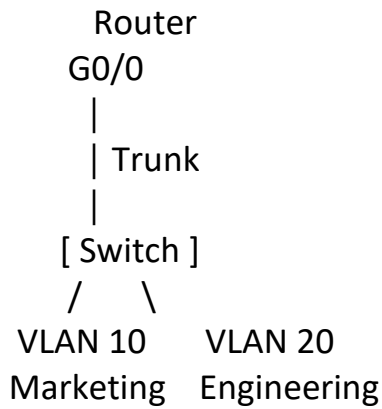
PC1 needs a **Layer-3 device** to move the packet between the two different networks.

That is the job of **inter-VLAN routing**.

Why Router-on-a-Stick?

Instead of adding another physical router interface, we can use **one physical router interface divided into multiple logical subinterfaces.**

Example:



On the router:

G0/0.10 → VLAN 10 → 192.168.10.1

G0/0.20 → VLAN 20 → 192.168.20.1

The switch sends VLAN-tagged traffic through the single trunk link, and the router's subinterfaces route traffic between the VLANs.

Main advantages

1. Saves hardware

You don't need a separate physical router interface for every VLAN.

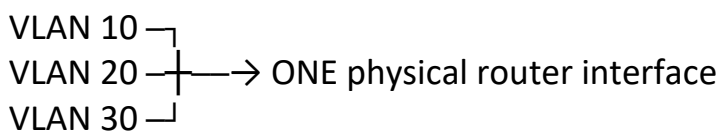
Traditional:

VLAN 10 → Physical Interface

VLAN 20 → Physical Interface

VLAN 30 → Physical Interface

With Router-on-a-Stick:



2. Lower cost

For a small startup, using one existing router avoids buying additional router hardware or interface modules.

3. Easy to expand

If the company adds another VLAN, you can create another subinterface rather than installing another physical interface.

Example:

G0/0.10 → Marketing
G0/0.20 → Engineering
G0/0.30 → HR
G0/0.40 → Finance

4. Centralized configuration

All inter-VLAN routing can be configured on the same router.

Router-on-a-Stick vs Multiple Physical Interfaces

Router-on-a-Stick	Multiple physical interfaces
One physical interface	One physical interface per VLAN
Uses subinterfaces	Uses physical interfaces
Lower hardware cost	More hardware required
Easy to add VLANs	Limited by available interfaces
Requires trunk link	Separate links/interfaces
Suitable for small networks	Can provide better performance at higher traffic

One important disadvantage

Router-on-a-Stick has **one physical link carrying traffic for multiple VLANs**. Therefore, that link can become a bottleneck if the network has heavy inter-VLAN traffic.

For this **small startup with a single router**, however, Router-on-a-Stick provides a practical way to enable Marketing ↔ Engineering communication without adding another physical router interface.

“Inter-VLAN routing is required because VLANs are separate Layer-2 broadcast domains and a Layer-2 switch cannot route traffic between them. Router-on-a-Stick allows one physical router interface to provide routing for multiple VLANs using logical subinterfaces over a trunk link, reducing hardware cost and making the network easier to scale.”